Homework 3 Report

1. Problem 1:

1.1 Least-Squares Formulation

The goal is to find the optimal similarity transformation to map a set of control points from a moving image to a fixed image. A similarity transformation accounts for rotation, translation, and isotropic scaling.

Let (x, y) denote a point in the moving image and (x', y') denote the corresponding point in the fixed image. The transformation can be expressed as:

$$x' = a * x - b * y + tx$$
$$y' = b * x + a * y + ty$$

where a and b incorporate both scaling and rotation such that:

$$a = s * \cos(angle)$$
$$b = s * \sin(angle)$$

The error for a single point pair is modeled as:

$$e_i = X_i * \theta - f_i$$

The variables are defined as:

- **f_i (Target Vector):** This contains the known coordinates of the i-th control point in the fixed image: f_i = [x'_i; y'_i]
- **θ (Parameter Vector):** This contains the unknown transformation parameters: θ = [a; b; tx; ty]
- **X_i (Data Matrix):** This matrix organizes the coordinates of the i-th control point from the moving image:

  X_i = [x_i, -y_i, 1, 0; y_i, x_i, 0, 1]

By stacking all point correspondences, the system can be written as:

$$X * \theta = f$$

Since the system is overdetermined, the transformation parameters are estimated in the least-squares sense by solving the stacked system X * theta = f using MATLAB's backslash operator:

```
theta = X \ f
```

Algorithm:

1. **Parameter Estimation:** Construct a global matrix X by stacking all X_i matrices and a global vector f by stacking all f_i vectors. The transformation parameters θ are estimated using the least-squares solution.
2. **Output Initialization:** A new output image is initialized with the same dimensions as the moving image (per the assignment, the fixed image is assumed to have the same size, so the output dimensions are consistent).
3. **Inverse Spatial Mapping:** To avoid holes in the transformed image, inverse mapping is used. For each pixel location in the output image, the corresponding coordinate in the moving image is computed using the inverse of the similarity transformation.
4. **Interpolation:** Since the computed source coordinates may not fall exactly on integer pixel locations, nearest-neighbor interpolation is used. The intensity value is assigned from the closest pixel in the moving image.
5. **Boundary Handling:** If the mapped coordinate lies outside the valid range of the moving image, the pixel remains at the initialized gray value.

1.2 Results and Discussion

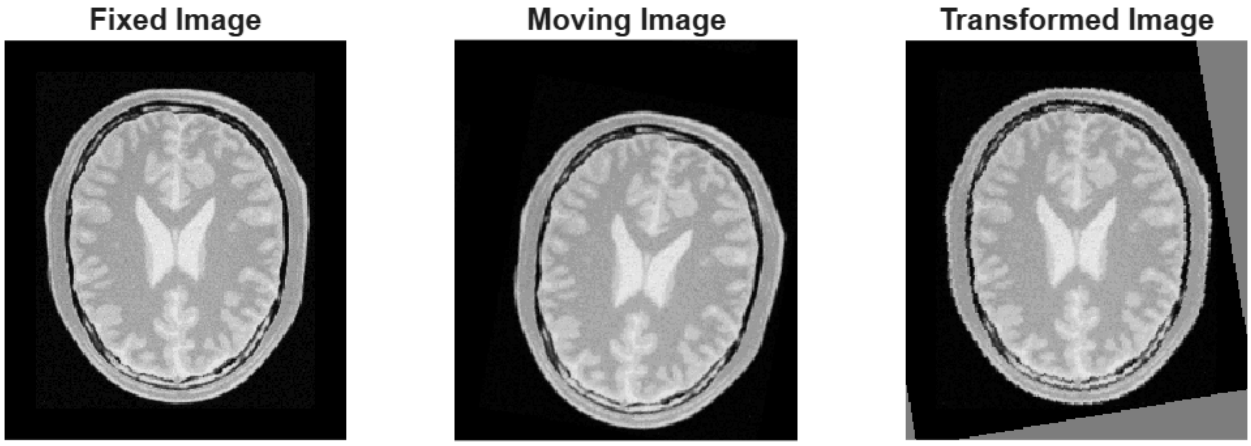


Figure 1: Registration of brain proton density slices. The moving image is transformed and aligned into the spatial coordinates of the fixed image.
The transformed image shows that the moving brain slice was mapped into the coordinate system of the fixed image using the estimated similarity transformation. The ventricles, outer skull boundary, and overall head shape align closely after transformation, indicating that the estimated parameters are consistent with the selected point correspondences. Because the output image was initialized to gray and inverse mapping was used, pixels outside the valid source region remain gray rather than producing holes in the transformed image.

The accuracy of the registration was verified using the following methods:

- **Visual Comparison:** The transformed image closely aligns with the fixed image, with minimal visible structural differences in the overlapping regions.
- **Mathematical Verification:** Substituting the moving image control points into the estimated transformation equations produces coordinates that closely match the corresponding fixed image points, indicating a low alignment error.

Correctness was evaluated in two ways. First, visual comparison indicates that the transformed image overlaps the fixed image closely in the main anatomical structures. Second, the transformed control points were compared with the corresponding fixed-image points. The average alignment error was approximately 0.64 pixels, which is small relative to the image size and indicates that the transformed control points remain close to their corresponding fixed-image points. This result shows that the similarity transformation is suitable for aligning the two images when reliable point correspondences are used.

2. Problem 2:

The image Fig1118(a).tif contains yeast cells with bright white blobs inside many of the cell bodies. Since the assignment defines the cells as the gray regions rather than the internal white blobs, the main objective of preprocessing is to suppress the bright internal structures while preserving the overall cell shapes. This allows the segmentation stage to focus on the actual cell bodies instead of being misled by the bright spots.

2.1 Watershed

The watershed-based method was applied after a foreground mask of the cell bodies was obtained and refined. Since the bright white blobs inside the cells are not part of the target regions, preprocessing was first used to suppress these internal structures while preserving the gray cell bodies. The binary mask was then refined by hole filling, small-object removal, opening, and closing. After that, the distance transform was computed so that the center of each cell had a larger value than the boundary region. Height-based minimum suppression was applied to the negative distance map before watershed in order to reduce over-segmentation caused by small local minima. The separated regions were then labeled and measured to compute the cell areas and average area.

Algorithm:

1. **Preprocessing:** Convert the image to grayscale, apply Gaussian smoothing, and suppress the bright white blobs using morphological opening.

2. **Foreground Mask Generation:** Generate a binary mask using thresholding, then refine it by filling holes, removing small objects, and applying opening and closing.
3. **Distance Transform:** Compute the distance transform of the mask so that the center of each cell is emphasized.
4. **Watershed Segmentation:** Apply height-based minimum suppression (imhmin with threshold 1.5) to the negative distance transform, then apply the watershed transform to separate touching cells.
5. **Final Labeling:** Remove watershed ridge lines, label the remaining connected regions, and treat them as individual cells.
6. **Area Measurement:** Measure the area of each labeled cell and compute the average cell area.

2.2 Region Growing

The region-growing method was also applied on the foreground mask so that segmentation remained restricted to likely cell regions rather than spreading into the background. Seed points were obtained from regional maxima of the distance transform and reduced to compact markers so that each seed represented a likely cell center. During growth, 8-connected neighbors were examined, and a candidate pixel was added only if it belonged to the foreground mask, had not already been labeled, and satisfied the required intensity constraints. This made the growth process more constrained than simple thresholding and limited leakage across adjacent cells. Any remaining unlabeled pixels inside the mask were assigned to the nearest grown region, and holes were filled so that the final labels represented solid cell bodies more consistently.

Algorithm:

1. **Preprocessing:** Convert the image to grayscale, suppress the bright white blobs using morphological opening, and generate a foreground mask using thresholding, hole filling, small-object removal, and morphological opening and closing.
2. **Seed Initialization:** Compute the distance transform of the foreground mask and extract regional maxima as seed points. Reduce each seed to a compact marker and assign a unique label to each one.
3. **Region Growth:** Grow each labeled seed using 8-connected neighbors. A candidate pixel is added only if it belongs to the foreground mask, has not already been labeled, and satisfies the intensity constraints.
4. **Remaining Pixel Assignment:** Assign any unlabeled pixels inside the foreground mask to the nearest grown region so that the full cell body is covered.
5. **Hole Filling:** Fill holes inside each labeled region so that the segmented result represents solid cell bodies rather than partial interiors.

6. **Area Measurement:** Measure the area of each labeled cell and compute the average cell area.

2.3 Comparison of the Two Methods

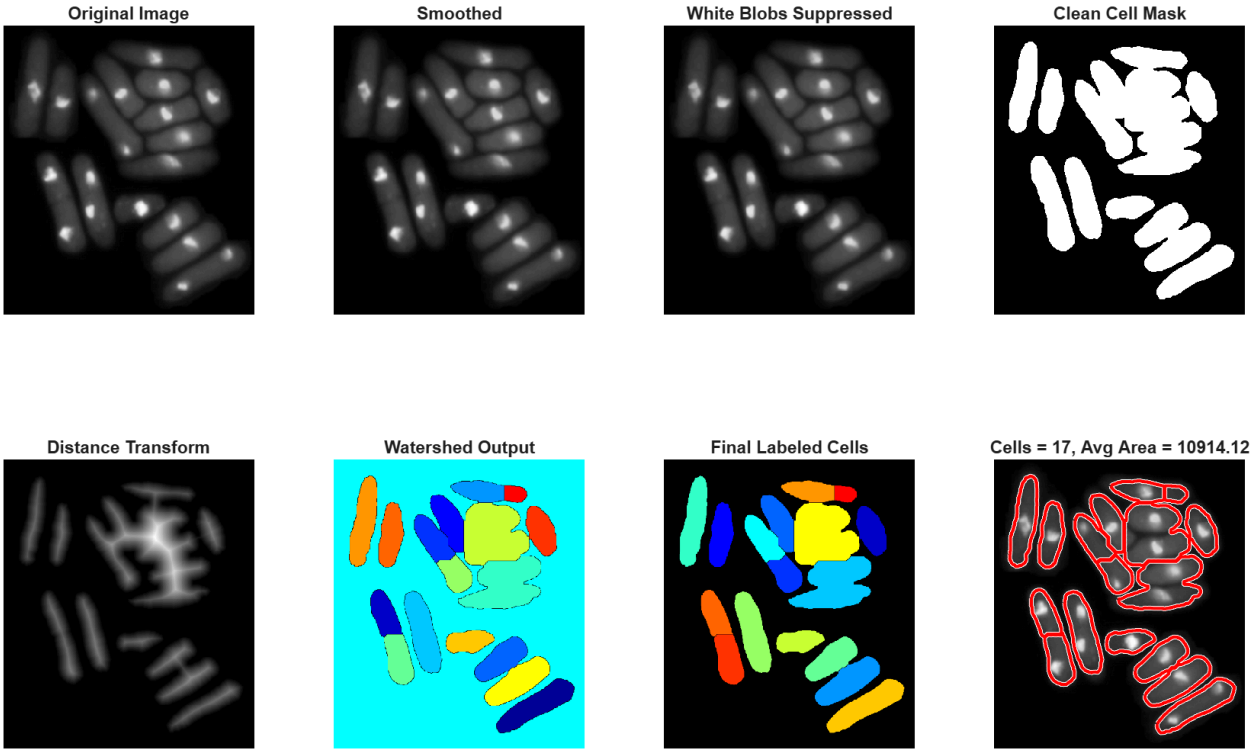


Figure 2: Watershed-based segmentation of yeast cells. The figure shows the original image, smoothed image, white-blobs-suppressed image, clean cell mask, distance transform, watershed output, final labeled cells, and boundary overlay.

In Figure 2, the watershed-based method uses the foreground mask and distance transform to separate touching cells. These steps display that the preprocessing stage produces a mask for watershed segmentation, while the distance transform emphasizes the central regions of the cells before splitting connected objects. The outcome detects 17 cells with an average area of 10914.12 pixels. The segmented regions cover most of the visible cell bodies, and the boundaries between neighboring cells are separated in this result.

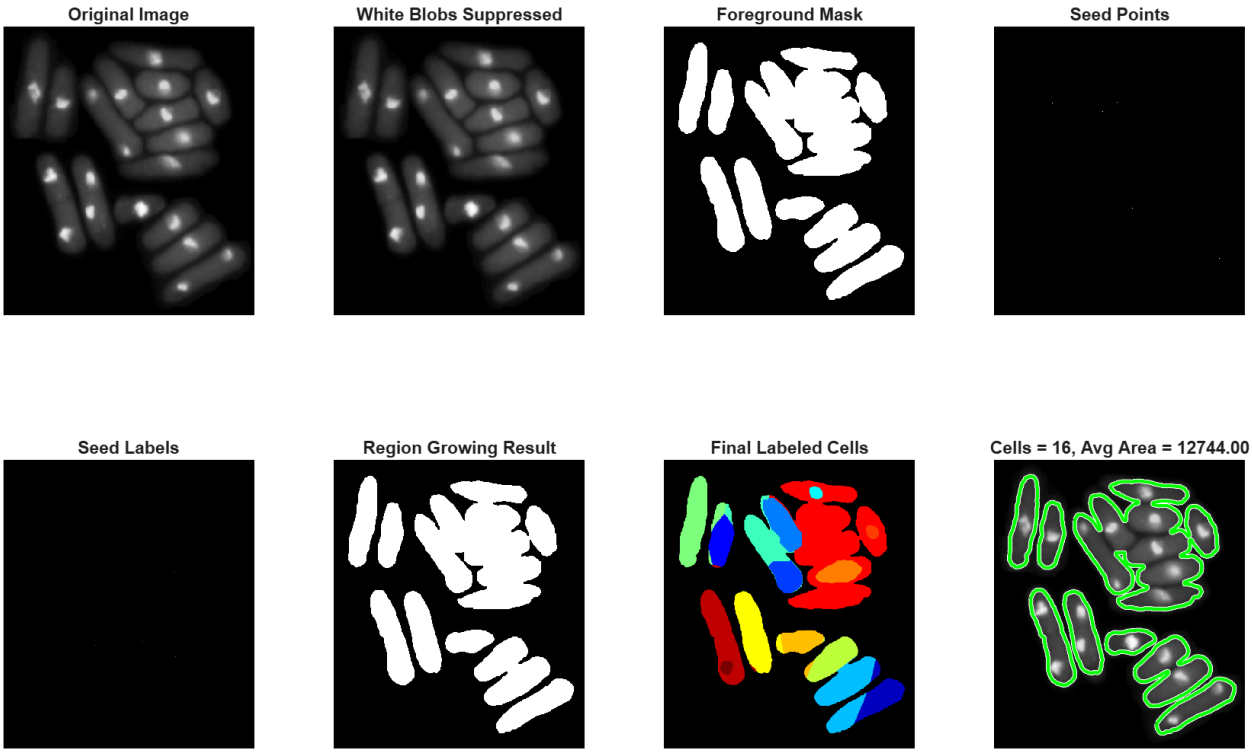


Figure 3: Region-growing segmentation of yeast cells. The figure shows the original image, white-blobs-suppressed image, foreground mask, seed points, seed labels, region-growing result, final labeled cells, and boundary overlay.

In Figure 3, the region-growing method begins from seed points extracted from the foreground mask and expands each seed into a cell region based on local intensity similarity. The results suggest that the preprocessing stage removes the influence of the bright internal blobs and restricts the growth process to likely cell regions. The result detects 16 cells with an average area of 12744.00 pixels. Although many of the cell bodies are recovered, the result remains more sensitive to seed placement and growth constraints, and some touching regions are not separated as cleanly as in the watershed result.

Comparison Result:

Both methods depend on preprocessing to suppress the bright white blobs and preserve the gray cell bodies. The region-growing method provides a valid segmentation approach, but its result is more sensitive to the seed generation and growth conditions. In contrast, the watershed method separates touching cells more effectively after the distance transform is applied. Based on the outputs, the watershed method produces better separation between neighboring cells in this image and gives a consistent result for area measurement. The region-growing method may be an alternative, but for this dataset the watershed result is clearer in the touching-cell regions.

3. Resources that Helped Me

MATLAB Documentation:

MATLAB Left Matrix Division (mldivide)
https://www.mathworks.com/help/matlab/ref/double.mldivide.html
MATLAB Morphological Operations
https://www.mathworks.com/help/images/morphological-filtering.html
MATLAB Image Segmentation
https://www.mathworks.com/help/images/image-segmentation.html
MATLAB Global Image Threshold Using Otsu's Method (graythresh)
https://www.mathworks.com/help/images/ref/graythresh.html
MATLAB Distance Transform of Binary Image (bwdist)
https://www.mathworks.com/help/images/ref/bwdist.html
MATLAB Watershed Transform
https://www.mathworks.com/help/images/ref/watershed.html
MATLAB Extended-Maxima Transform (imextendedmax)
https://www.mathworks.com/help/images/ref/imextendedmax.html
MATLAB Binarize Image (imbinarize)
https://www.mathworks.com/help/images/ref/imbinarize.html

Others:

A Threshold Selection Method from Gray-Level Histograms
https://engineering.purdue.edu/kak/computervision/ECE661.08/OTSU_paper.pdf
Image Segmentation
https://disp.ee.ntu.edu.tw/class/tutorial/Segmentation%20tutorial.pdf
Image Registration Lecture Notes
https://user.engineering.uiowa.edu/~aip/lectures/ch8_w_highlights.pdf

Course Material:

Course lecture notes